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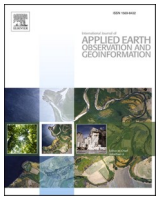

Large-scale high-resolution coastal subsidence mapping in eastern China with Sentinel-1 and Sentinel-2: Heterogeneous patterns and primary drivers☆

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ABSTRACT

A majority of the low-lying coastal areas worldwide, where the population is densely concentrated, are confronted with high to extremely high risks of land subsidence. However, a comprehensive detection and quantification of large-scale coastal subsidence patterns and their primary drivers in eastern China remain lacking. In this study, we employed Time Series Interferometric Synthetic Aperture Radar (TS-InSAR) technique and Sentinel-1 data to derive the vertical deformation pattern at a resolution of 90 m from 2017 to 2024. We developed a novel multi-frame mosaicking method, achieving spatially consistent InSAR observations over the land-sea transition areas. Our findings uncover extensive coastal subsidence in northeastern Shandong and eastern Jiangsu, highlighting several rapid-subsidence funnels with rates exceeding 50 mm/yr for the first time. By integrating Sentinel-2 multispectral imagery, subsidence time series, groundwater level measurements, and principal component analysis (PCA), we further analyzed the spatiotemporal distribution patterns and underlying drivers of these heterogeneous subsidence funnels. Our analysis demonstrates that anthropogenic factors are the dominant drivers of coastal subsidence. Four representative case studies reveal distinct subsidence mechanisms: (1) brine extraction for salt production and aquaculture, (2) excessive freshwater withdrawal for agricultural irrigation and industrial use, (3) groundwater depletion for intensive greenhouse aquaculture, and (4) land reclamation for industrial infrastructure development. In each region, subsidence patterns are predominantly controlled by a single dominant factor. This study is expected to provide valuable insights for monitoring and managing coastal subsidence, enhance our understanding of associated risks, and offer critical guidance for protecting communities in vulnerable coastal areas.

1. Introduction

In the coastal lowlands around the world, there are numerous megacities with a population of over 10 million. These low-lying coastal areas are highly sensitive to global climate change and human activities (Kulp and Strauss, 2019; Nicholls and Shirzaei, 2024; Reimann et al., 2023). The rapid population growth has significantly increased the demand for natural resources, further deepening our reliance upon, and transformative impacts on, coastal environments (Neumann et al., 2015). Land subsidence is a significant challenge in coastal low-lying

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zones, causing issues such as ground rupture (Peng et al., 2016), flooding inundation (Li et al., 2023; Nicholls et al., 2021; Shirzaei and Bürgmann, 2018), coastline erosion (Gambolati et al., 1999), relative sea level rise (Tay et al., 2022), reduced aquifer storage capacity (Hasan et al., 2023), and infrastructure damage (Nicholls et al., 2021), all of which pose disasters and potential risks to both natural ecosystems and human communities (Bagheri-Gavkosh et al., 2021; Herrera-García et al., 2021; Wang et al., 2024b).

The drivers of coastal subsidence can be categorized into natural processes and human activities. The former includes tectonic movement, earthquake cycle, glacial isostatic adjustment, and the consolidation and compaction of surface sediments (Escalona and Mann, 2011; Karegar et al., 2016; Meltzner et al., 2006; Ren et al., 2024; Wolstencroft et al., 2014). The latter encompasses the mining, extraction of subsurface fluids (e.g., water, oil, and gas), infrastructure construction (e.g., subways, tunnels), and other load-generating activities (Azadnejad et al., 2024; Hu et al., 2022; Jones et al., 2016; Minderhoud et al., 2018; Motagh et al., 2017; Nikos et al., 2016; Sun et al., 2023; van der Horst et al., 2018; Wang et al., 2022a). Among the various causes of subsidence, natural processes typically occur over long-time scales and result in relatively modest magnitudes of subsidence. Currently, numerous studies have indicated that the extraction of subsurface fluids, particularly groundwater withdrawal, is the primary factor driving coastal subsidence, especially densely populated and agriculturally developed areas (Bagheri-Gavkosh et al., 2021; Huning et al., 2024; Shirzaei et al., 2021; Syvitski et al., 2009).

Time Series Interferometric Synthetic Aperture Radar (TS-InSAR) is an efficient space geodetic technique that derives surface deformation by performing differential interferometry on SAR images acquired at different times (Osmanoğlu et al., 2016). It has wide coverage, all-weather and all-time capabilities, and high accuracy in the deformation detection and monitoring. Traditional TS-InSAR data processing is primarily designed for single-frame analysis. However, when the study area covers a large region spanning multiple tracks and frames, multi-frame mosaicking becomes necessary. The uneven distribution of land and sea in coastal areas and the spatial heterogeneity of subsidence further complicate regional bias estimation (Li et al., 2019; Wang et al., 2024c; Yunjun et al., 2019).

Existing multi-frame mosaicking methods can generally be divided into two categories: (1) mosaicking the SAR raw data or intermediate products (Casu et al., 2008; Haghshenas Haghighi and Motagh, 2024; Tong et al., 2018), and (2) mosaicking the TS-InSAR deformation results (Du et al., 2023; Hussain et al., 2018; Kalia et al., 2017). The former typically yields better spatial consistency across large areas but is limited to data from the same orbit and demands a huge amount of computational resources. The latter focuses on estimating and correcting systematic biases between deformation results from different frames, by using high-coherence points, tie points, regional average offsets, or through the integration of external datasets such as GNSS observations (Ou et al., 2022; Shen et al., 2024; Sun et al., 2017; Wang et al., 2012; Wang et al., 2024c). However, this approach faces several challenges, such as the imaging geometry of SAR systems, which often leads to degraded data quality near frame edges. In addition, variations in observation geometries, reference point selection, error distribution across frames, and inconsistencies can affect the accuracy and reliability of mosaicked results. Therefore, these factors collectively make it highly challenging to obtain accurate mesurements of large-scale coastal subsidence.

Most areas along the eastern coast of China are facing high to very high risks of potential subsidence (Fang et al., 2022; Herrera-García et al., 2021). The spatiotemporal characteristics of subsidence in Tianjin and Shanghai have been extensively studied, with human interventions significantly mitigating and controlling subsidences (Su et al., 2023; Wang et al., 2022b; Yin et al., 2015). However, our understanding of long-term spatiotemporal subsidence pattern and primary drivers in Shandong and Jiangsu provinces remains limited. Therefore, mapping large-scale high-resolution subsidence in these regions is an urgent priority.

In this study, we first apply TS-InSAR techniques using Sentinel-1 imagery to generate accurate high-resolution vertical deformation rate maps for the eastern China. Second, we develop a multi-frame mosaicing method specifically tailored for coastal areas. Third, we integrate PCA method, Sentinel-2 imagery and land cover data to characterize subsidence patterns for different areas. In addition, we analyze the correlation between groundwater level and subsidence. This study aims to provide a foundational reference for coastal subsidence monitoring, disaster management, and sustainable development.

2. Study area and datasets

2.1. Study area

We focus on the Shandong and Jiangsu provinces in the eastern China (Fig. 1), which have large populations and developed marine economies but still lack high spatiotemporal resolution InSAR observations, as the study areas. (Herrera-García et al., 2021; Huning et al., 2024).

Shandong Province is located at the lower reaches of the Yellow River, while the central mountainous area rises up. As the highest point, Mount Tai has an altitude of 1532.7 m. The Yellow River Delta (YRD) has an average altitude of less than 10 m and is the lowest land. Shandong Peninsula protrudes into the Bohai Sea and the Yellow Sea, accounting for 1/6 of the country's total coastline, boasting a mainland coastline of 3,505 km (Li et al., 2024a). At the end of 2023, the permanent resident population of Shandong Province exceeded 100 million, with the gross regional product surpassing 9 trillion yuan.

Jiangsu Province lies in the lower reaches of the Yangtze River and Huai River, with the Yellow Sea to the east. It spans a land area of 107,200 km^2, predominantly consisting of plains, thereby facilitating advanced agricultural development. Jiangsu's coastline extends approximately 954 km, with a total coastal wetland area of 3819 km^2, representing about 1/4 of the national total. In 2023, its gross regional product surpassed 12 trillion yuan, while the permanent resident population reached 85.26 million.

2.2. Datasets

Fig. 1 presents the spatial distribution of the SAR observations, GNSS stations, and groundwater wells used in this study. **Table S1** provides detailed information of Sentinel-1 SAR data, including track and frame numbers, acquisition time spans, and incidence angles. The SAR images consist of 7 tracks and 16 frames with time period covering from March 2017 to December 2023. We obtained the Sentinel-1 raw data in single-look-complex (SLC) format and interferometric wide swath (IWS) mode, which are freely available from the Alaska Satellite Facility (https://search.asf.alaska.edu/) and cover a 250 km swath with a spatial resolution of 5 m × 20 m. The revisit cycle is 12 days, which ensures at least two observations within a month. The total number of original SLC images is 2,486, and the data volume amounts to 10 terabytes. Sentinel-1B ended its mission on December 23, 2021 due to a power supply failure and Sentinel-1A provided almost no descending orbit data over the study area. Therefore, we primarily used ascending orbit data from the Sentinel-1A.

Furthermore, we collected raw GNSS data from nine continuous stations (Fig. 1 and **Table S2**) in the Crustal Movement Observation Network of China (CMONOC) for the period 2010–2023. The raw data are processed using GAMIT/GLOBK software version 10.7 for InSAR accuracy evalution (Wang et al., 2024a). The final GNSS products can be freely available from GNSS data product service platform of China Earthquake Administration (http://www.cgps.ac.cn). They are referenced to the World Geodetic System (WGS84) ellipsoid, and are consistent to the height datum of InSAR and digital elevation model

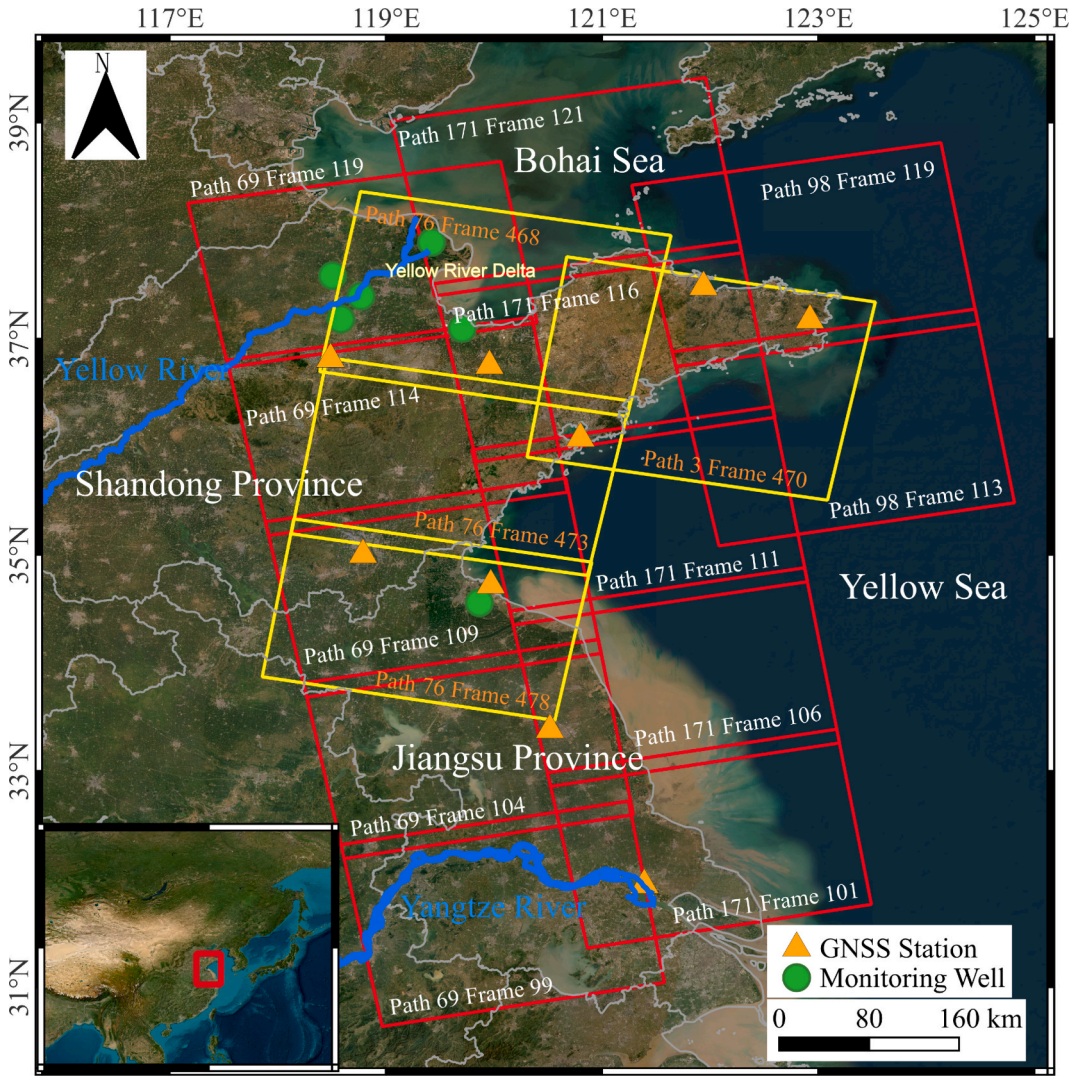


Fig. 1. Location of Shandong and Jiangsu provinces, along with the Sentinel-1 data coverage over the study area. Red and yellow rectangles indicate the ascending and descending SAR frames, respectively. Orange triangles indicate the CMONOC GNSS stations, while green circles represent the monitoring wells.

(DEM) data (Li et al., 2013). The average uncertainties of the GNSS-derived deformation rates are superior to 1 mm/yr.

We also collected the groundwater data from the China Geological Environment Monitoring Groundwater Level Yearbook compiled by the China Geological Environment Monitoring Institute (Fig. 1 and **Table S3**). Due to the systematic offsets in well station information before and after 2018, we used groundwater level data from 2018 to 2021. The data were recorded at five-day intervals, resulting in six observations per month and a total of 72 observations annually. However, some periods contain missing data due to various on-site observational challenges.

In addition, we used Sentinel-2 optical images at both ends of the InSAR time span to analyze the changes in land cover in the typical subsidence area. For the areas with more significant changes, additional Sentinel-2 images are selected for further analysis and discussion. The Sentinel-2 datasets are freely available from Copernicus Data Space Ecosystem (https://browser.dataspace.copernicus.eu). The spatial resolution is 10 m, and there are 13 spectral bands, including visible light (VIS), near-infrared (NIR), and short-wave infrared (SWIR).

As shown in Fig. 2, the 30 m Landsat-derived annual land use land cover (LULC) product of China freely available (https://zenodo.org/records/15853565) was incorporated to investigate the correlation between subsidence and land cover (Yang and Huang, 2021). The 1:5,000,000 geologic chronogram of surface outcrops from the United States Geological Survey (https://catalog.data.gov/dataset/generalized-geology-of-southeast-asia-geo3bl) was employed to support the subsidence evaluation (Steinshouer et al., 1999).

3. Methods

3.1. TS-InSAR processing

As shown in Fig. 3, we firstly used the InSAR Scientific Computing Environment (ISCE) developed by NASA to generate interferograms from individual scenes of SLC data (Fattahi et al., 2017). Subsequently, the interograms were stitched rather than simultaneous processing of multiple SLC images. The original pixel resolution of SLC is 2.3 m in the range direction by 13.9 m in the azimuth direction. When generating the interferogram, we performed 20x6 multi-look processing on the SLC to mitigate noise, enhance coherence, and improve processing efficiency. Interferograms were generated by pairing each acquisition with the five preceding and five subsequent acquisitions, resulting in over 900 interferograms for each frame under different paths (**Fig. S1**). The 30 m NASADEM released by NASA Land Processes Distributed Active Archive Center in 2020 was used to remove topographic phases, co-register and geocode, which can be freely available from NASA EarthData platform

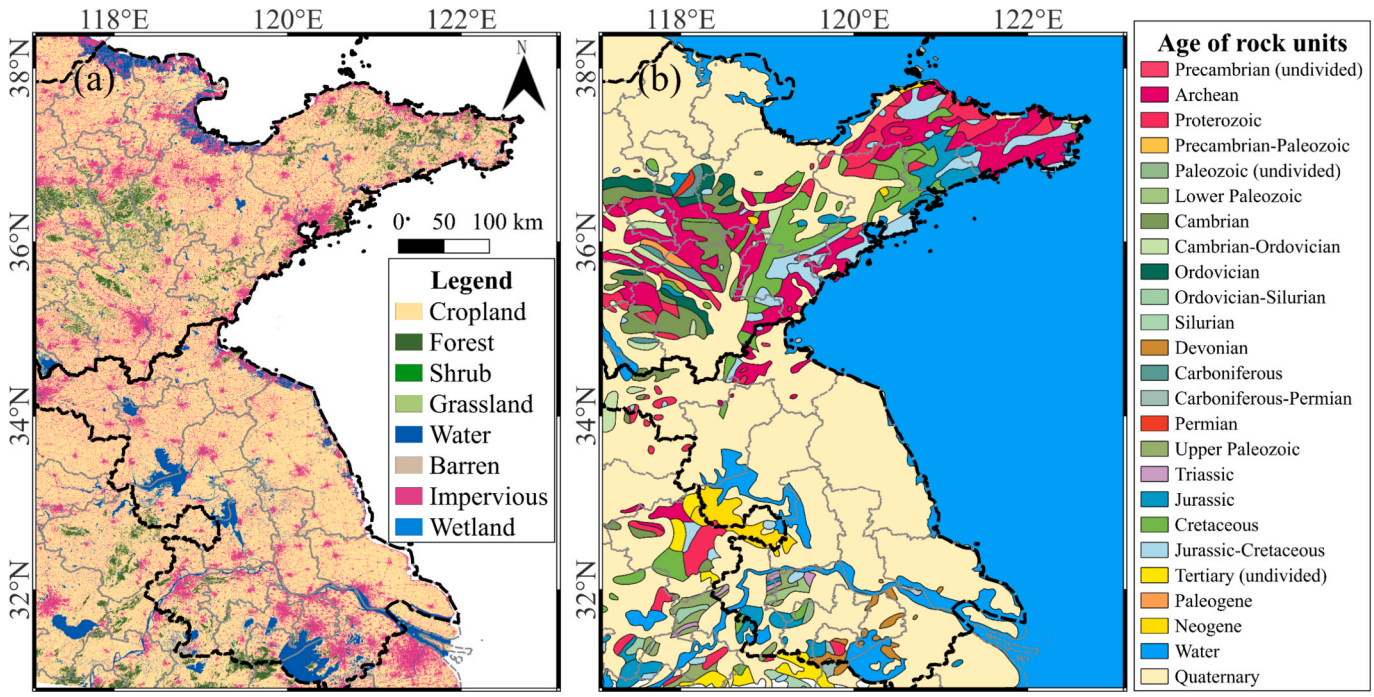


Fig. 2. LULC map and geologic chronogram of surface outcrops over the study area.

(https://search.earthdata.nasa.gov). The statistical-cost network-flow algorithm for phase unwrapping (SNAPHU) was used for phase unwrapping (Chen and Zebker, 2001). Finally, we resampled the final geocoded deformation result from 84 m to 90 m.

We then used the Miami INsar Time-series software in PYthon (MintPy) to implement the Small Baseline Subset (SBAS) time series analysis (Berardino et al., 2002; Yunjun et al., 2019) with the stack interferograms produced by the ISCE and estimated the ground surface displacement in line-of-sight (LOS) direction. In the standard workflow of the MintPy, the interferogram stack was initially inverted to derive the raw phase time-series. Subsequently, deterministic phase components such as the tropospheric delay, the topographic residual, and phase ramp were corrected. This process generated a noise-reduced displacement time-series. Following this, the average velocity was estimated, excluding noisy SAR acquisitions. These noisy acquisitions were identified through an outlier detection method based on the root mean square (RMS) of the residual phase (Yunjun et al., 2019). In this study, we used a coherence threshold of 0.7 to constrain the network of interferograms. This workflow calculated the average spatial coherence for each interferogram. A minimum spanning tree (MST) network was then formed with the inverse of the average coherence as the weight. Interferograms were excluded if their average coherence was below the threshold and they were not part of the MST network (Chaussard et al., 2015).

Specifically, we used the bridging and phase closure methods to correct unwrapping errors based on the specific conditions of each SLC image region. We also calculated and projected the solid earth tides (SET) displacement in the horizontal and vertical directions to the LOS direction according to the 2010 International Earth Rotation and Reference System Service (IERS) (Yunjun et al., 2022). We estimated and mitigated tropospheric delays in the interferograms through the generic atmospheric correction online service (GACOS) (Yu et al., 2017). Due to the short spatial wavelength of the land subsidence signal, we then estimated and removed residual tropospheric delay, ionospheric delay, and orbital errors for each acquisition based on the reliable pixels (Yunjun et al., 2019). The systematic topographic phase residual resulting from the DEM error was estimated by its proportionality to the perpendicular baseline time-series (Fattahi and Amelung, 2013). As shown in **Fig. S2**, unwrapped phase interferograms before and after correction from different paths and frames were selected to demonstrate the effect of the error correction. The phase standard deviation (STD) of the radar interferograms is shown in **Fig. S3**. After applying the tropospheric correction using GACOS, the phase STD decreased for 77 % of the interferograms. After all corrections, 98.6 % of the interferograms showed a reduction in phase STD compared to the original interferograms. Finally, the average velocity in the WGS84 coordinate system was determined as the slope of the line that best fits the displacement time-series, which has been adjusted after applying the aforementioned corrections.

3.2. Multi-frame splicing

To derive spatially consistent observations over large areas, we developed a multi-frame mosaicking method tailored for coastal areas, based on the theory of uncontrolled block adjustment (Pi et al., 2019; Wang et al., 2024c; Zhang et al., 2015). The principle of this method is that, observations in the overlapping regions between frames should be consistent. By minimizing the total difference between observations in overlapping areas, we constructed a set of bias correction equations that link the systematic bias and correction terms for each frame (Fig. 4).

Assuming the study area contains M InSAR frames with N overlapping regions, for the i-th overlapping region, the deformation observations of the subsidence rate in the two frames are defined as def_i^1 and def_i^2. Theoretically, the two values should be equal. Therefore, the objective function is to minimize the deviation between the values across all overlapping regions, as follows:

$$\min\left(\sum_{i=1}^{N}\left(\left|def_i^1 - def_i^2\right|\right)\right) \tag{1}$$

For all overlapping regions, the deviation correction equation can be expressed as below:

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix} = \underset{N\times M}{B} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} + \begin{bmatrix} \Delta_1 \\ \Delta_2 \\ \vdots \\ \Delta_N \end{bmatrix} \tag{2}$$

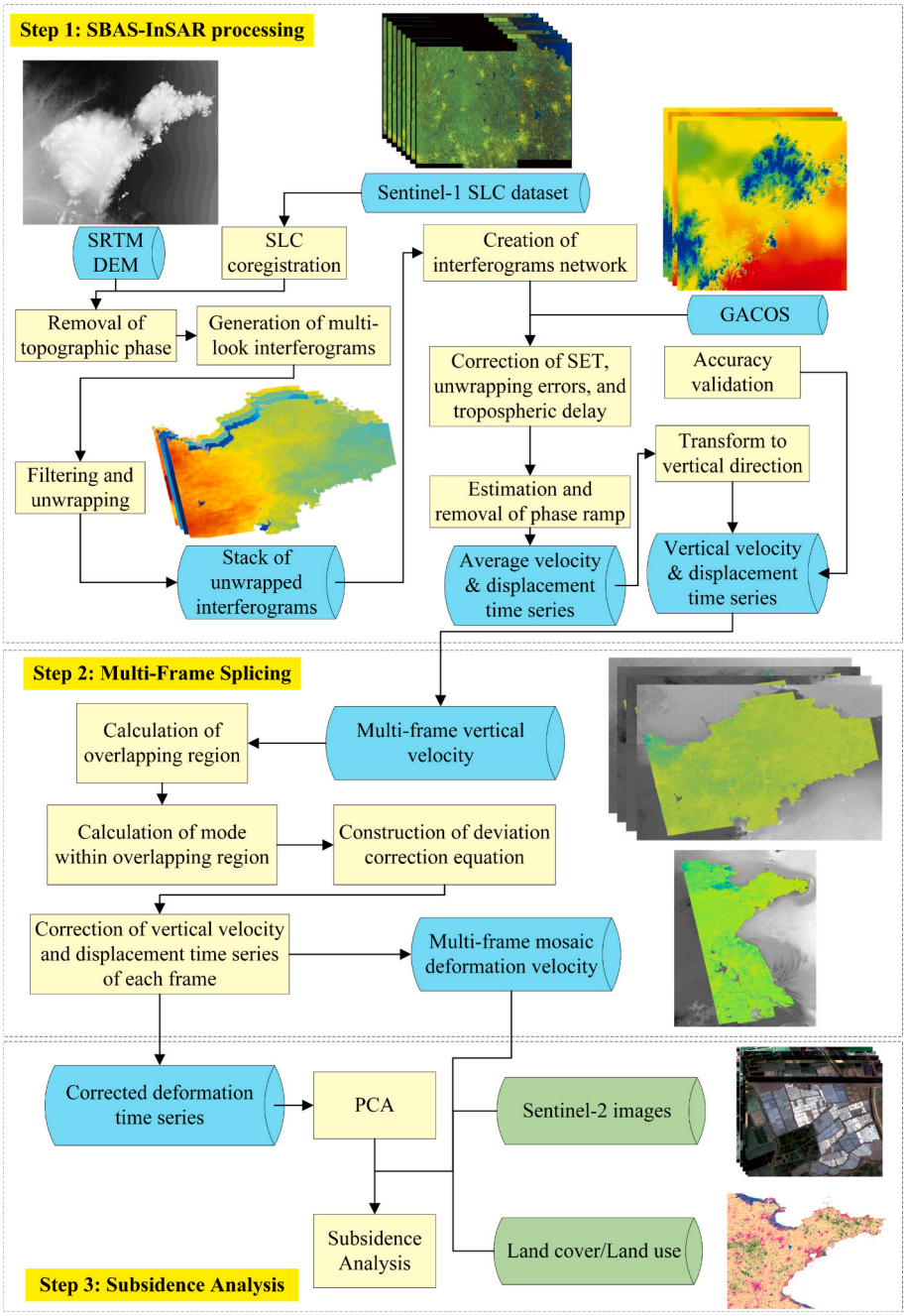


Fig. 3. Workflow of the data processing methods.

simplified as

$$V = B \cdot X + \Delta \tag{3}$$

where V is the deviation matrix, and each element v_i is defined as $\left| def_i^1 - def_i^2 \right|$.$B$ is an $N \times M$ coefficient matrix, where in each row, the elements corresponding to two SAR frames with overlapping areas are set to 1 and -1, respectively, while all other elements are set to 0. X is the matrix composed of correction values to be determined. Δ is the matrix of residual errors.

Then, a generalized weighted least-squares solver, based on the Moore-Penrose pseudoinverse, is used to calculate the correction value (CV) as below:

$$\widehat{X} = \left(B^T P B\right)^+ B^T P V \tag{4}$$

where $\left(B^T P B\right)^+$ denotes the Moore-Penrose pseudoinverse. $\boldsymbol{P}$ is a weight matrix where each element is determined based on the variance of each SAR frame and the area of its corresponding overlap region.

Since large-scale deformations typically occur only in localized areas, while the majority remain stable or exhibit small-amplitude deformations, using the mode as the observation value is more representative than using the mean or median (Costantini et al., 2017; Siting et al., 2014). However, the deformation rates are stored as single-precision floating-point numbers, and most of the them are unique values. Therfore, it is difficult to directly perform mode-based statistics. Furthermore, if there are extensive deformation signals in the overlapping area, a single mode cannot accurately represent the overall distribution of the data. To address the aforementioned issues, we achieved large-scale coastal subsidence monitoring through the multi-frame stitching of the land-sea boundary area (Fig. 3). The main procedure is as

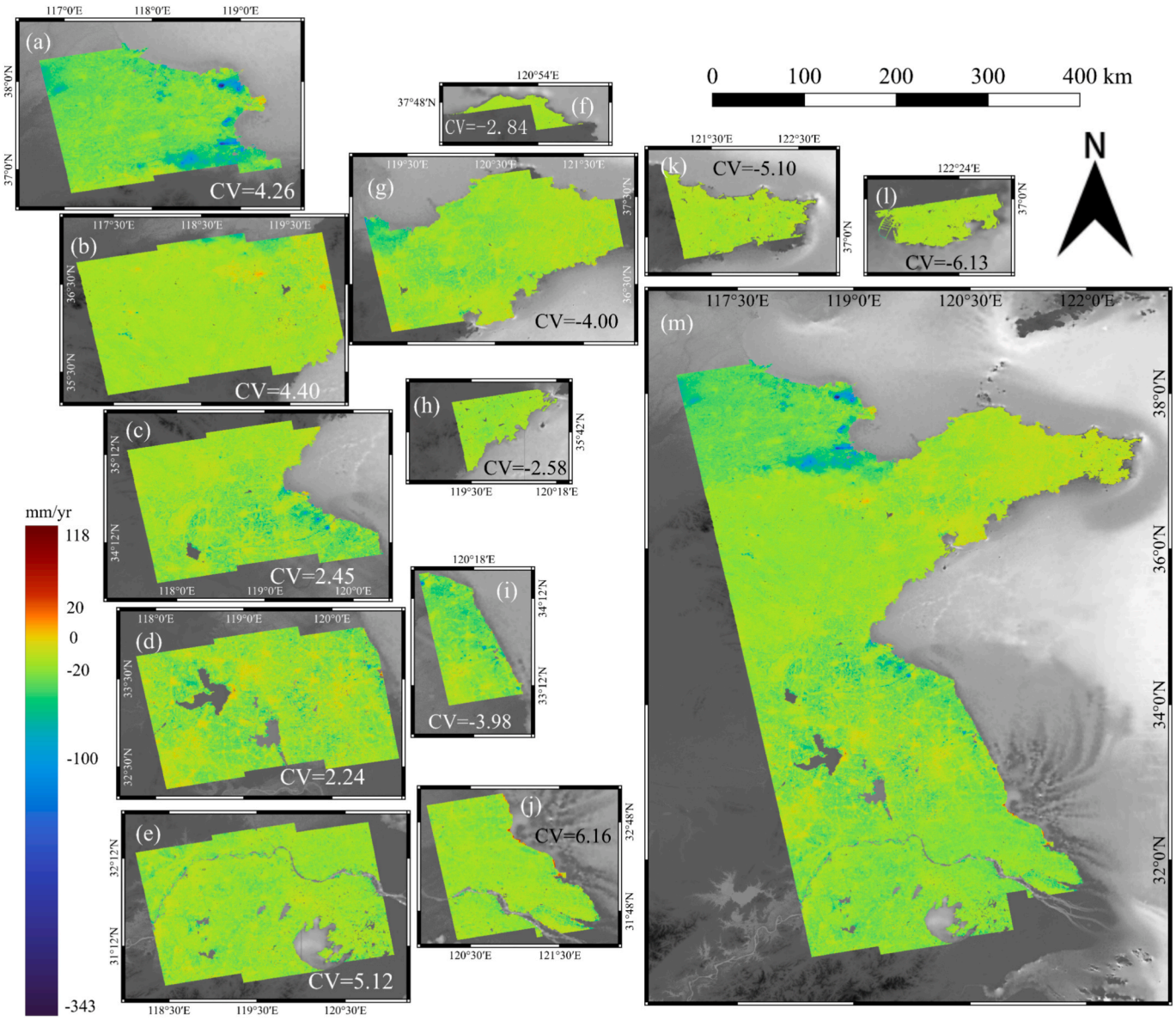


Fig. 4. Annual average vertical deformation rate maps with corresponding correction values (CV) for each InSAR frame (a-l) and the combined deformation rate map coverting the entire study area (m). The minus sign represents subsidence. The specific paths and frames correspond to: (a) P69F119, (b) P69F114, (c) P69F109, (d) P69F104, (e) P69F99, (f) P171F121, (g) P171F116, (h) P171F111, (i) P171F106, (j) P171F101, (k) P98F119, and (l) P98F113. The base map is the SRTM DEM.

follows.

(1) The projection of a three-dimensional displacement vector $\vec{d} = [d_v \quad d_n \quad d_e]^T$ onto LOS direction can be expressed as following (Erten et al., 2010; Hu et al., 2014):

$$\begin{cases} \vec{d}_r^{\,A} = \delta_r + d_v\cos\theta_{inc} - \sin\theta_{inc}[d_n\sin H + d_e\cos H] \\ \vec{d}_r^{\,D} = \delta_r + d_v\cos\theta_{inc} - \sin\theta_{inc}[d_n\sin H - d_e\cos H] \end{cases} \tag{5}$$

where θ_{inc} and H denote the incidence angle and the heading angle (positive clockwise from the North), respectively. $\vec{d} = [d_v \quad d_n \quad d_e]^T$ indicates the deformation vector in vertical, north and east direction, and the superscripts of A and D shows the ascending and descending geometry, respectively. Since there are three unknowns for the displacement from an ascending orbit, another interferometric observation from a descending orbit is required. However, due to the limited availability of descending orbit, only a portion of the study area is covered by both ascending and descending orbits. The horizontal deformation $\sqrt{\vec{d_n^2} + \vec{d_e^2}}$ in **Fig. S4** derived from ascending and descending orbits (Erten et al., 2010) indicates that the study area does not exhibit significant horizontal motion. This finding is also consistent with previous GNSS and InSAR studies (Guo et al., 2019; Peng et al., 2020; Wang and Shen, 2020; Wei et al., 2014), which similarly suggest the absence of notable horizontal displacement in the region. Due to the relatively small horizontal ground motion, the LOS deformation from the ascending orbit was then converted to vertical displacement using the radar incidence angle θ and the formula $\boldsymbol{d}_v = \boldsymbol{d}_{LOS}/\cos\theta$ (Chaussard et al., 2014b).

(2) After obtaining vertical deformation, automatic identification and extraction of overlapping regions between multi-frames from multi-tracks were performed. Then, the frequency distribution histogram of the deformation rate in the overlapping regions was used for mode statistics (**Fig. S5**). The top five frequency intervals were selected from the histogram, and the median of each

interval was determined. The average of these medians, which better reflects the overall condition of the region, was then used as the observation value to construct the deviation correction equations.

(3) The inverse of the average variance of the two frame forming the overlapping region was used as the weight to construct the weight matrix P. Since significant deformation may occur in the overlapping region, or the overlapping region may be located at the land–ocean boundary, where the amount of valid data is relatively limited, the weight matrix P needs to be adjusted based on the effective area and the magnitude of deformation in the overlapping region.

(4) Finally, after applying the aforementioned deviation corrections to all frames with overlapping areas, we obtained the rate maps as a unified result covering the entire study area, with values in overlapping regions representing the average of multiple SAR frames.

3.3. Deformtion characterization with PCA

We used Principal Component Analysis (PCA) to extract the subsidence patterns across different areas and analyze the correlations between subsidence and drivers. By performing linear combinations of the original features, PCA generates several components with the largest variance, known as principal components (PCs), which retain the most critical information from the original data. These PCs are mutually independent and uncorrelated, representing the primary directions of variation within the data (Bonì et al., 2016; Jolliffe and Cadima, 2016; Karimzadeh et al., 2018; Wang et al., 2022a). The main steps of PCA are as follows:

(1) Matrix S is constructed based on the deformation time series data and it is centralized to obtain S_C.

(2) The covariance matrix C is calculated using the following formula:

$$C = \frac{1}{n-1} S_C^T S_C \tag{6}$$

(3) Perform eigenvalue decomposition on the covariance matrix C to obtain the eigenvalues λ and their corresponding eigenvectors x.

(4) Select principal components based on the magnitude of the eigenvalues, where each PC has an associated variance explanation ratio (VER) that represents its contribution to the original data. The VER can be calculated as follows:

$$VER_i = \frac{\lambda_i}{\sum_{i=1}^{k} \lambda_i} \tag{7}$$

The PCA results primarily reveal different patterns of temporal variation in the time series and their contributions. However, it should be noted that the number of PCs obtained from PCA does not directly correspond to the number of drivers, nor does each PC necessarily represent a specific cause of subsidence. For land subsidence, the PC1 represents the feature with the largest data variance, corresponding to the overall trend of subsidence and reflecting the proportion of major controlling factors (Kang and Knight, 2023; Li et al., 2023; Wang et al., 2022a). The PC2 captures secondary variations, usually corresponding to periodic changes (Chaussard et al., 2014a; Wang et al., 2022a). The remaining PCs, based on their VER, may represent other influencing factors or noise-related error signals (Peng et al., 2022; Si et al., 2024).

4. Results

As shown in Fig. 4, vertical deformation rate maps at a resolution of 90 m for each SAR frame from 2017 to 2023 were obtained through SBAS-InSAR processing. Subsequently, using the multi-frame splicing method, those results from multiple frames were spliced to produce an average surface deformation rate map covering the entire study area. The CV annotated on the rate map for each InSAR frame involved in constructing the deviation correction equation was calculated from Equation (4) and applied to the deformation rate correction.

To better study the coastal subsidence, we used the latest coastline datasets from the OpenStreetMap (OSM) (https://osmdata.openstreetmap.de/data/coastlines.html) to build the coastal buffer area by extending landward to a distance 100 km from shore (Reid et al., 2005). We consider the deformation rate range of ± 10 mm/year as a relatively stable interval, that is, the nominal or zero subsidence class (Hasan et al., 2023; Shirzaei et al., 2021). This interval represents the upper limit of most of the natural processes of vertical land motion, e.g., glacial isostatic adjustment, tectonics, surface deposition and compaction, and observation uncertainty. As shown in Fig. 5, large-scale subsidence is mainly distributed in the northwestern part of Shandong Province and the eastern part of Jiangsu Province with the strata primarily consisting of Quaternary deposits, while those mountainous and hilly areas are relatively stable. In the northwestern part of the Shandong Province, 47.4 % of the area experiences an annual average subsidence rate of less than 20 mm/year, 7.4 % is less than 50 mm/year, and 0.9 % is less than 100 mm/year. In contrast, the majority (83.1 %) of the Jiangsu coastal plains has deformation rates within the range of [-20, 20] mm/year, with approximately 16.6 % of the area experiencing subsidence rates more than 20 mm/year.

We compared the GNSS vertical deformation rates from nine continuously operating stations with the InSAR results in Fig. 6. For each GNSS station, we identified the nearest high-coherence InSAR pixel, which is straightforward and computationally simple (Delen et al., 2024; Yan et al., 2024). However, it may be highly sensitive to noise in a single InSAR pixel. It assumes the GNSS station and the InSAR pixel are measuring the exact same deformation, which may not be accurate if there are localized variations. The root mean squre (RMS) error between the two datasets is 1.3 mm/year, indicating a good consistence. The boxplot further demonstrates that the errors exhibit a small dispersion. It is important to note that GNSS stations are typically located in relatively stable areas where InSAR has undergone multi-look processing. In addition, we also compared the deformation time series of GNSS with that of InSAR. As shown in Fig. S7, the overall performance of these two independent observations shows good consistency in deformation. Observation disruptions occurred at two GNSS stations. Meanwhile, in the regions with severe decorrelation, InSAR observations are affected by potential phase unwrapping errors, resulting in discrete outliers, such as at the JSNT station.

Furthermore, the Sentinel-1 deformation results from both ascending and descending orbits were used for cross comparison to evaluate the reliability of the SBAS-InSAR results.We employed robust measures to account for the non-normality and outliers from the observation errors (Li et al., 2021). **Fig. S7** shows that a total of 968,498 overlapping pixels from the ascending (Path 69 Frame 119) and descending (Path 76 Frame 468) data were used for accuracy evaluation. The correlation coefficient between the ascending and descending results is 0.97, indicating a strong positive correlation. The mean difference is 1.0 mm/yr, with the STD value of 6.9 mm/yr and RMS value of 7.0 mm/yr. The median difference is 0.8 mm/yr, the median absolute deviation (MAD) is 4.3 mm/yr, the normalized median absolute deviation (NMAD) is 6.3 mm/yr, and the absolute deviation at 90 % quantile (Le90) is 9.8 mm/yr. The STDs for the remaining pairs of ascending and descending data are 6.9 mm/yr, 3.86 mm/yr, and 6.14 mm/yr, respectively.

To further analyze the subsidence patterns of different subsidence funnels based on the PCA results, the temporal variations of each PC along with their corresponding eigenvalues were plotted, as shown in Fig. 7. The temporal variation trends of PC1 are highly consistent with the overall trends of the time series curves, indicating that PCA has successfully extracted the primary mode of variation in these areas. Furthermore, the VERs of PC1 are 94.7 %, 98.4 %, 99.0 %, 98.6 %, and

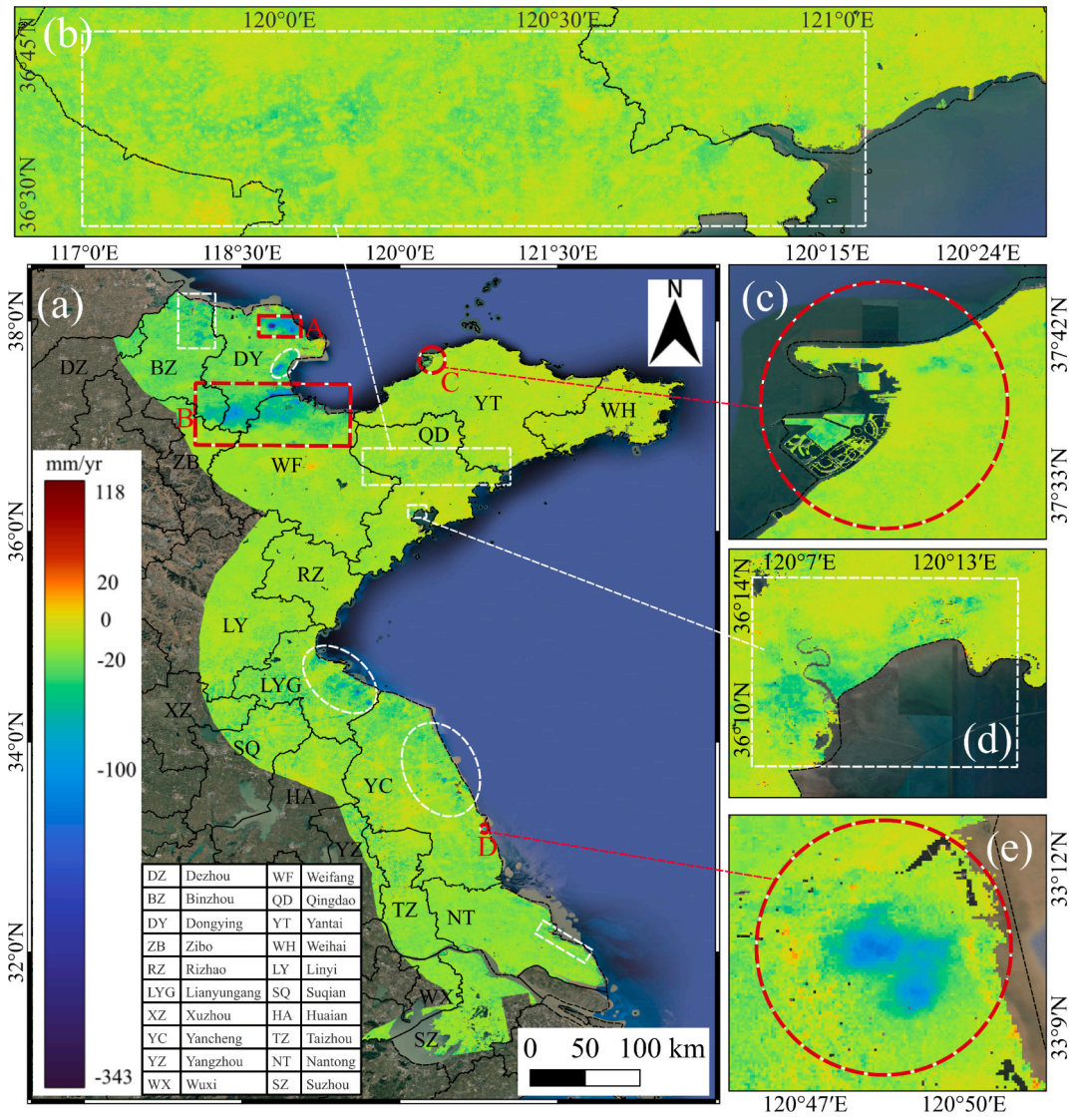


Fig. 5. Spatial distribution maps of coastal subsidence for (a) the entire study region and (b)-(e) selected subsidence zones. Subzones A-D marked in red rectangle and circles are highlighted for further discussion. White dashed lines indicate areas of concentrated subsidence. The base map is a Google Map satellite image, and black lines represent city-level administrative boundaries. The list of acronyms in the lower left corner provides abbreviated names for most coastal cities.

85.9 %, respectively, further demonstrating that the PC1 effectively captures the dominant variation characteristics of the regions. Therefore, the temporal variation of PC1 can represent the subsidence patterns in these areas. In contrast, the remaining PCs account for a very limited proportion of the variance and exhibit no clear temporal trends, making further analysis difficult.

5. Discussion

5.1. Primary drivers for typical subsidence zones

In this section, we conducted a detailed analysis on the spatiotemporal patterns and drivers of typical subsidence areas (Fig. 5 **and** Fig. S8). Taking into account different geographical locations and different driving factors causing subsidence, we selected four representative sub-regions (Zones A, B, C, and D), where the annual average coastal subsidence rate was all greater than 20 mm/year.

Zone A in Fig. 8 located in the Dongying City is a coastal alluvial plain of loose Quaternary sediments, forming a typical fluvial-dominated delta created by the river's high sediment load. Its hydrogeological context involves extensive porous aquifers and groundwater influenced by human activities like irrigation, affecting salinity and connectivity. It has the maximum subsidence rate of 340 mm/yr in the entire study area. 72.9 % of the area has a subsidence rate greater than 20 mm/year, while the region with a subsidence rate exceeding 50 mm/year covers 370 km^2, accounting for 39.7 % of the total subsidence area. In Fig. 8**(c)-(f)** and **Fig. S9**, it is evident that the most severely affected areas are located near the salt fields and fish ponds by visual interpretation from Sentinel-2 multispectral images and filed survey. This finding is further supported by the subsidence time series curves, which clearly show that the region has been experiencing severe ground subsidence with the magnitude of subsidence continuously increasing since March 2017. In the Z1 area, most points have experienced near-linear, continuous subsidence, which is still ongoing, except for the period from August 2019 to March 2020. In the Z2 area, the majority of points underwent significant subsidence in the earlier period before 2020, but subsidence has since tended to stabilize. The polynomial fitting curves effectively reflect this subsidence trend in the region. As shown in Fig. 8 **(h)-(j)**, the variance explanation ratio (VER) of PC1 reached 94.7 % and 98.4 % for Z1 and Z2, respectively, which indicates that the subsidence pattern in this area is predominantly controlled by a single driving factor. Therefore, the primary cause of coastal subsidence in this area is

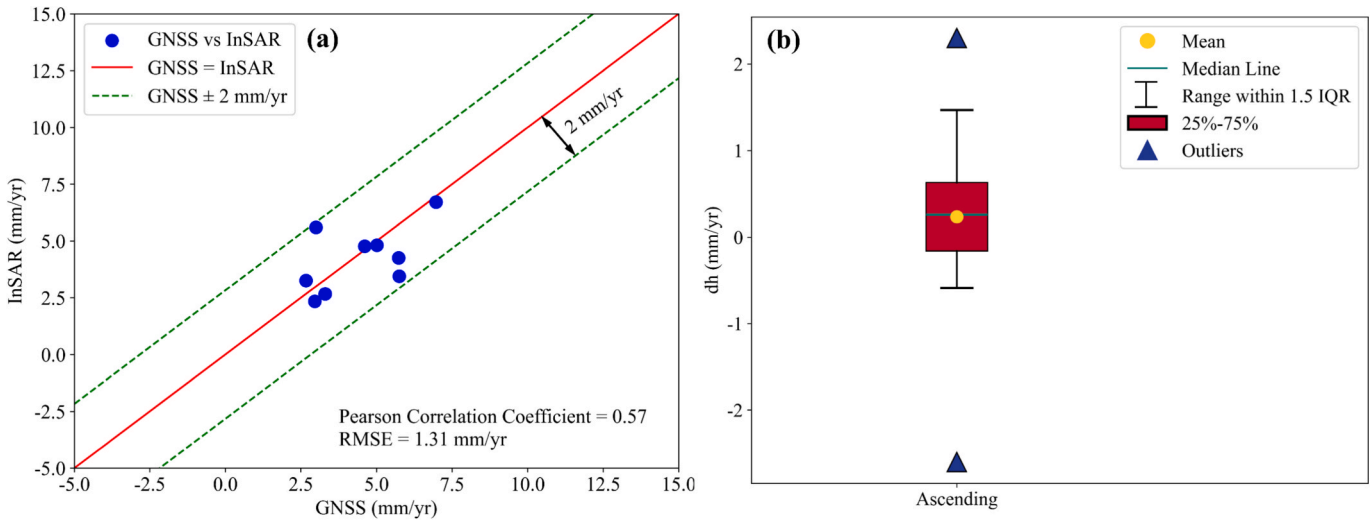


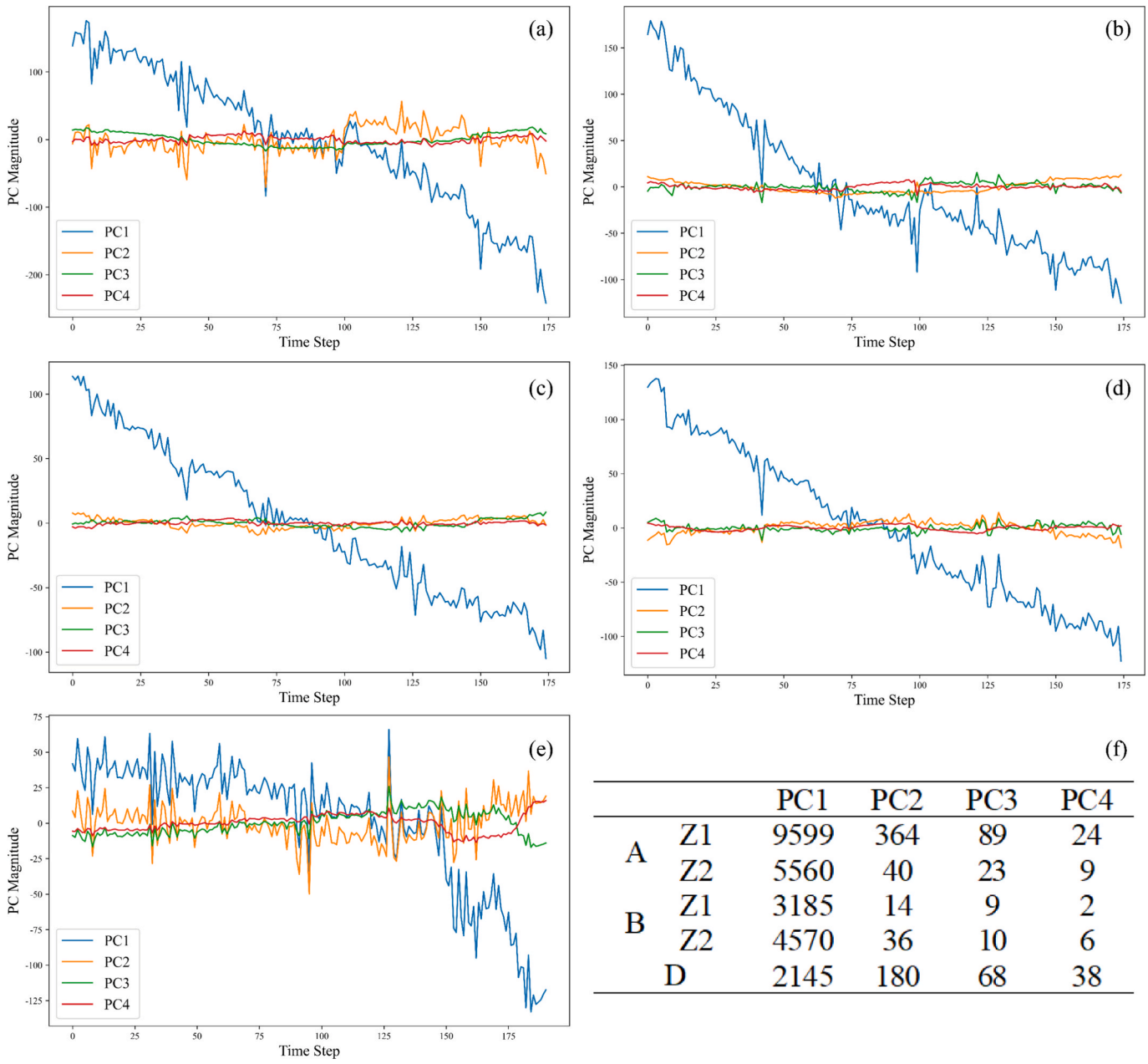
Fig. 6. Accuracy evaluation of vertical InSAR deformation rates compared with continuous GNSS measurements. (a) and (b) represent the scatter plot and box plot, respectively.

the extraction of brine for salt production and aquaculture purposes. The YRD, formed by the accumulation of sediments carried by the Yellow River, features relatively loose surface deposits that are prone to compaction (Li et al., 2024b). The extensive extraction of underground brine has accelerated the process of ground subsidence, further exacerbating the subsidence situation (Wang et al., 2022a).

Zone B in Fig. 9 located in the Laizhou Bay, has a complex geological history of transgression and regression, influenced by multiple riverine sediment sources, including the Yellow River. The primary aquifers in this area are composed of unconsolidated Quaternary sediments, which can be up to 300 m thick in the northern part. This geology, combined with extensive human activities, results in a hydrogeological system defined by pervasive salinization, complex groundwater–seawater mixing, and significant subsidence. It has the largest spatial extent of subsidence, exhibiting a maximum subsidence rate of 245 mm/yr and a maximum cumulative subsidence of 161.7 cm. The area where the subsidence rate exceeds 20 mm/year covers 5031 km^2, accounting for 57.5 % of the total area, with most of the subsidence rates concentrated within the range of 20 to 100 mm/year. Z1 and Z2, are located in Guangrao County, where the predominant land cover type consists of cropland, artificial structures, and salt fields (**Fig. S8**). Several petrochemical companies operate, primarily engaged in crude oil refining and the production of chemical products. The salt industry has a long history, with underground brine extraction beginning in the 1960 s. The history of subsidence dates back several years, with areas experiencing cumulative subsidence exceeding 300 mm from 2002 to 2008 (Liu and Huang, 2013). By 2019, there were over 1,600 brine wells, each with an extraction rate of approximately 100 m^3/day, and the total annual extraction volume for the entire region reached 3.20×10^7 m^3 (Wang and Zhu, 2022). The subsidence time series curves for Z1 and Z2 exhibit similar near-linear variation, indicating that subsidence has not been effectively controlled. Both Z1 and Z2 are dominated by a single factor, with the VER of the PC1 being 99.0 % for Z1 and 98.6 % for Z2. The primary driver causing subsidence is the extraction of shallow underground brine and deep groundwater used for agricultural irrigation, industrial production, and residential needs.

Zone C in Fig. 10 is located in the Yantai City. Its hydrogeology is defined by a complex, multi-layered aquifer system within the Quaternary sediments. These aquifers are highly vulnerable to seawater intrusion due to their coastal location and the intense water demands of the port and petrochemical complex. The extent and magnitude of subsidence are relatively small, but the subsidence is concentrated near Longkou Port and Yulong Island and exhibits distinct pattern with most areas being artificially land reclaimed coastal regions (**Fig. S9**). The area with an annual average subsidence rate greater than 20 mm/year accounts for 19.7 %. The maximum subsidence rate recorded is 92 mm/year with a maximum cumulative subsidence of 52.3 cm. Since around 2011, artificial land reclamation has been implemented on the Yulong Island, and the newly created land now hosts a significant petrochemical refining industrial base. Therefore, the extensive industrial facility construction has further intensified surface load stress, exacerbating the consolidation effects. It should not be overlooked, as subsidence poses potential risks to newly constructed industrial facilities. Similar subsidence patterns have been observed in Tongzhou Bay and Maojia Port in Nantong City, Shawan Port in Yancheng City, and Lianyungang Port in Lianyungang City, all caused by land reclamation and the construction of industrial facilities. Additionally, significant subsidence has been detected along the coastline of the Jiangsu Province, primarily attributed to land reclamation for aquaculture. On a global scale, the reclaimed coastal areas often experience subsidence that typically continues for several decades, as the placement of uncompacted fill over unconsolidated shallow seabed sediments provides potential for consolidation-induced subsidence (Sun et al., 2018).

Zone D in Fig. 11 located near the Wanggang River, Yancheng City, is low-lying coastal plain shaped by dynamic sedimentary and hydrological processes over millennia. Its hydrogeology is a product of shifting river courses and repeated marine transgressions, leading to a complex multi-layered system of unconsolidated sediments. Dominated by the porous aquifer system, it is characterized by widespread salinization, a dense surface water network, and land subsidence. It has a maximum subsidence rate reaching 138 mm/year. The area where the subsidence rate exceeds 20 mm/year accounts for 73.3 %, with the maximum cumulative subsidence reaching 118.9 cm. The significant subsidence began around May 2022, while prior to this period, the subsidence was minimal. However, after March 2023, the subsidence underwent notable changes once again. The maximum annual subsidence rate reached 630 mm/yr, with most areas (77.2 %) exhibiting subsidence rates exceeding 50 mm/yr. Sentinel-2 imagery highlight those areas with noticeable changes, that is, the total area of change is 13.0 km^2, representing 73.4 % of Zone D. We found that establishing greenhouse farms for shrimp cultivation has become a significant development in aquaculture in the Jiangsu Province in recent years. Therefore, land use has primarily shifted from traditional aquaculture ponds to greenhouse aquaculture facilities for the artificial farming of Pacific white shrimp. They are widely distributed in regions such as Qidong City, Rugao City, and Haian City in Nantong, Sheyang County, Dafeng District, and Binhai County in



(f)

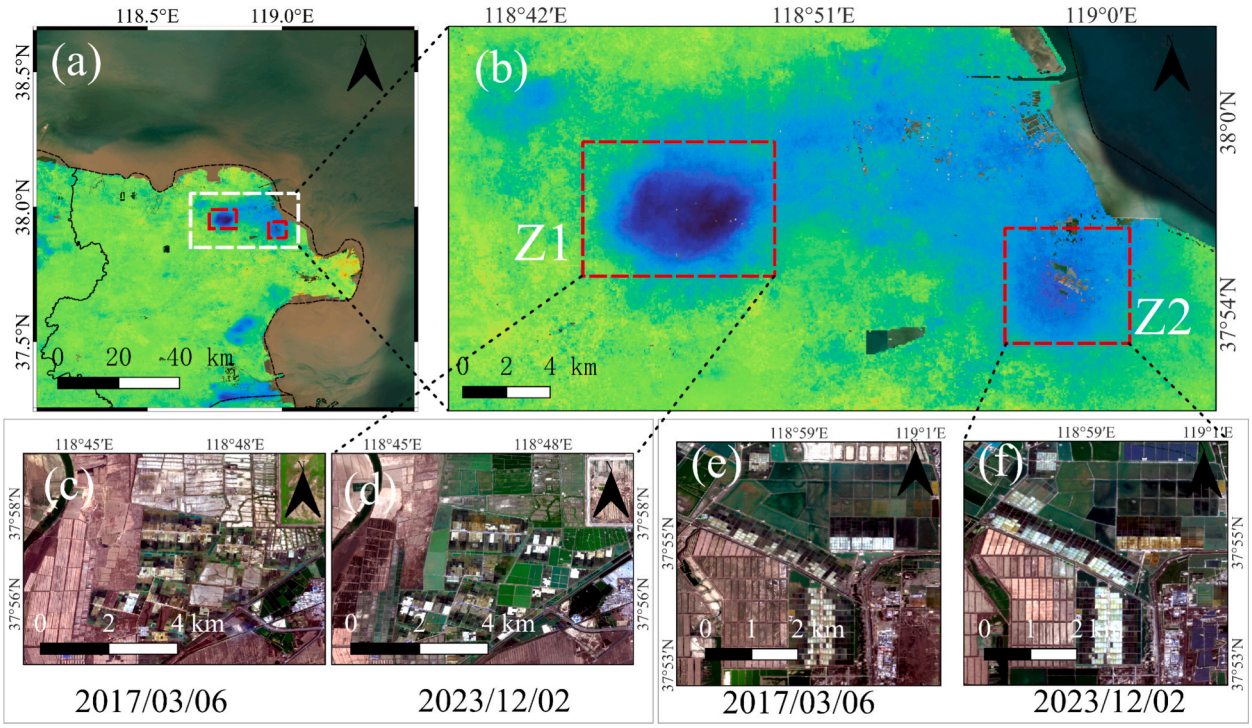
		PC1	PC2	PC3	PC4
A	Z1	9599	364	89	24
	Z2	5560	40	23	9
B	Z1	3185	14	9	2
	Z2	4570	36	10	6
D		2145	180	68	38

Fig. 7. Subplots (a)–(e) show the temporal trends of each PC, corresponding to Z1 and Z2 of Zone A, Z1 and Z2 of Zone B, and Zone D, respectively, while (f) presents the corresponding eigenvalues.

Yancheng, as well as Ganyu County in Lianyungang. Shrimp farming can typically use either seawater or groundwater. However, seawater contains bacteria and pathogens that pose significant threats to shrimp survival, negatively impacting their survival rates. In contrast, groundwater does not present these issues, making it the preferred choice for most greenhouse shrimp farmers. The spatial distribution of the significant subsidence areas closely aligns with the locations of the shrimp farming facilities, while the temporal changes of subsidence are strongly correlated with the development of these farms. Besides, the VER of PC1 is 85.9 %. Therefore, the subsidence is primarily due to the large-scale extraction of groundwater for shrimp farming.

By combining Sentinel-1 TS-InSAR subsidence monitoring with Sentinel-2 optical imagery, it is efficient to detect the distribution of subsidence zones and the development of shrimp farming facilities over a large area. Meanwhile, some local governments have recognized this issue. For example, the Water Resources Bureau of Yancheng City, has issued a notice stating that it will not approve water extraction applications from shrimp farmers within its jurisdiction and will investigate unauthorized well drilling and groundwater extraction by farmers. Similarly, the Nantong City has strengthened regulations regarding permits for newly established shrimp farming facilities and is taking significant measures to rectify illegal construction and groundwater extraction practices. The subsidence problem in this area still requires continuous monitoring in the future.

In addition, there are also some discretely-distributed weak uplift phenomena in the newly born land area of river estuary, as well as the silty salt marsh and tidal flats along the coast. This is consistent with previous studies (Ba et al., 2024; Cahoon, 2024; Ganju et al., 2024; Glover et al., 2023; Li et al., 2023), mainly due to the effect of salt marsh vegetation on the high turbidity water sediment interception, resulting in rapid sediment deposition and elevation of the ground.



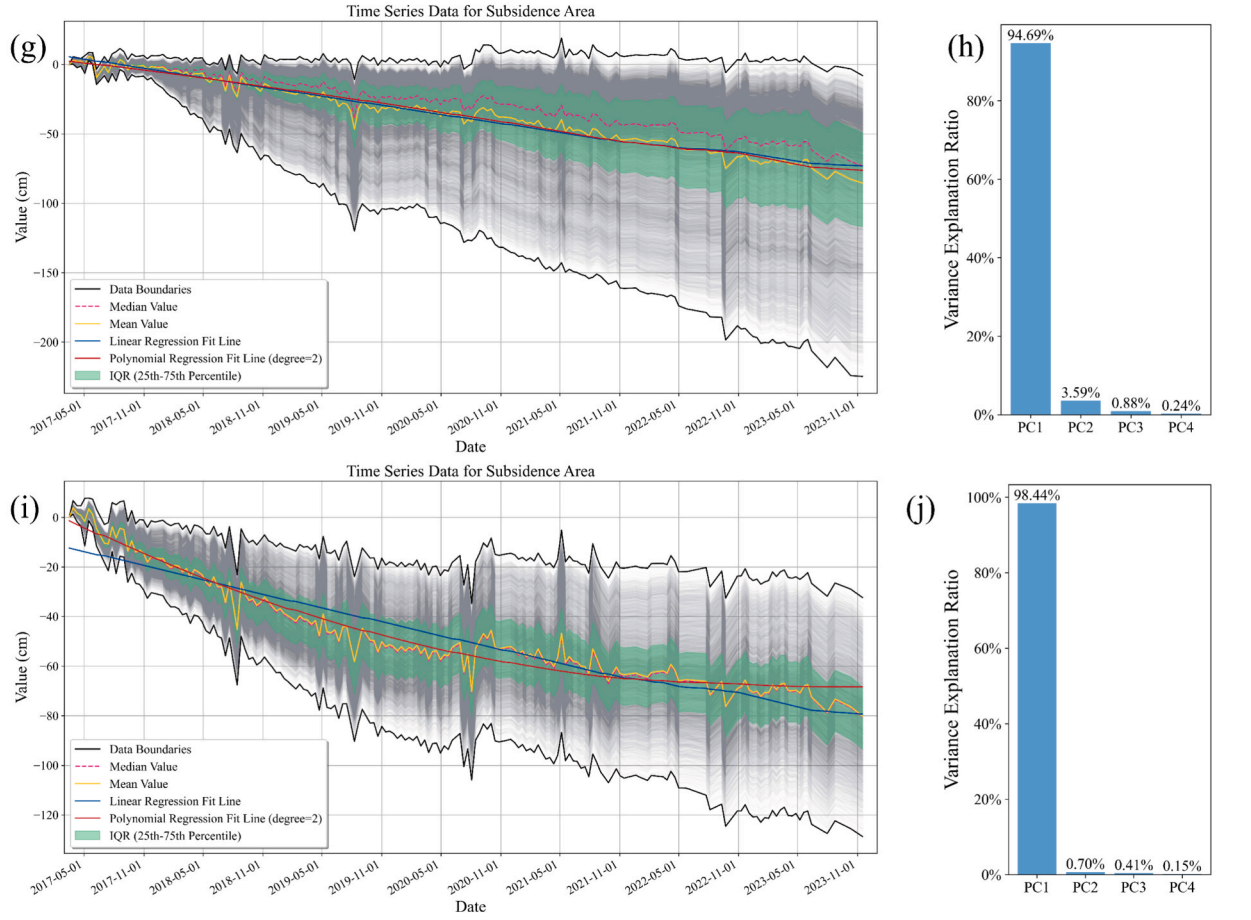


Fig. 8. (a) Spatial distribution of subsidence for Zone A. (b) shows the central area of subsidence funnels and is represented by a white dashed rectangle in (a). (c)-(f) consist of Sentinel-2 multispectral images for two subsidence funnels (Z1 and Z2), acquired at different times between March 2017 and December 2023. (g) and (h) present the surface deformation time series curves and PCA results derived from all pixels of Z1, while subplots (i) and (j) display corresponding results for Z2.

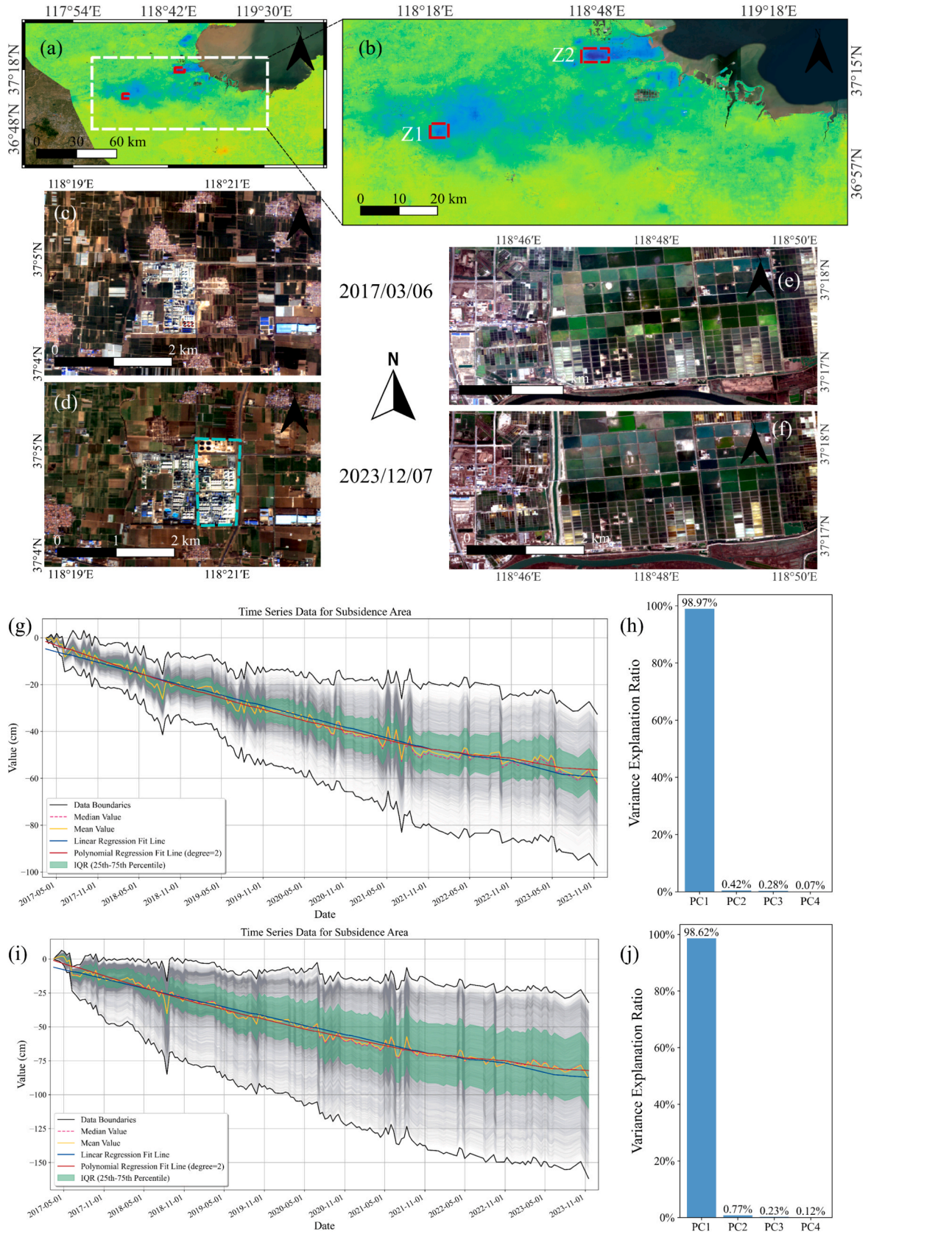


Fig. 9. (a)–(j) illustrate the spatial distribution of coastal subsidence for Zone B, along with the Sentinel-2 optical images, time series curves derived from all pixels of subsidence center (Z1 and Z2), and corresponding PCA results, consistent with those shown in Fig. 11. The green dashed polygon in the optical image of (d) represents the area with significant changes in surface features.

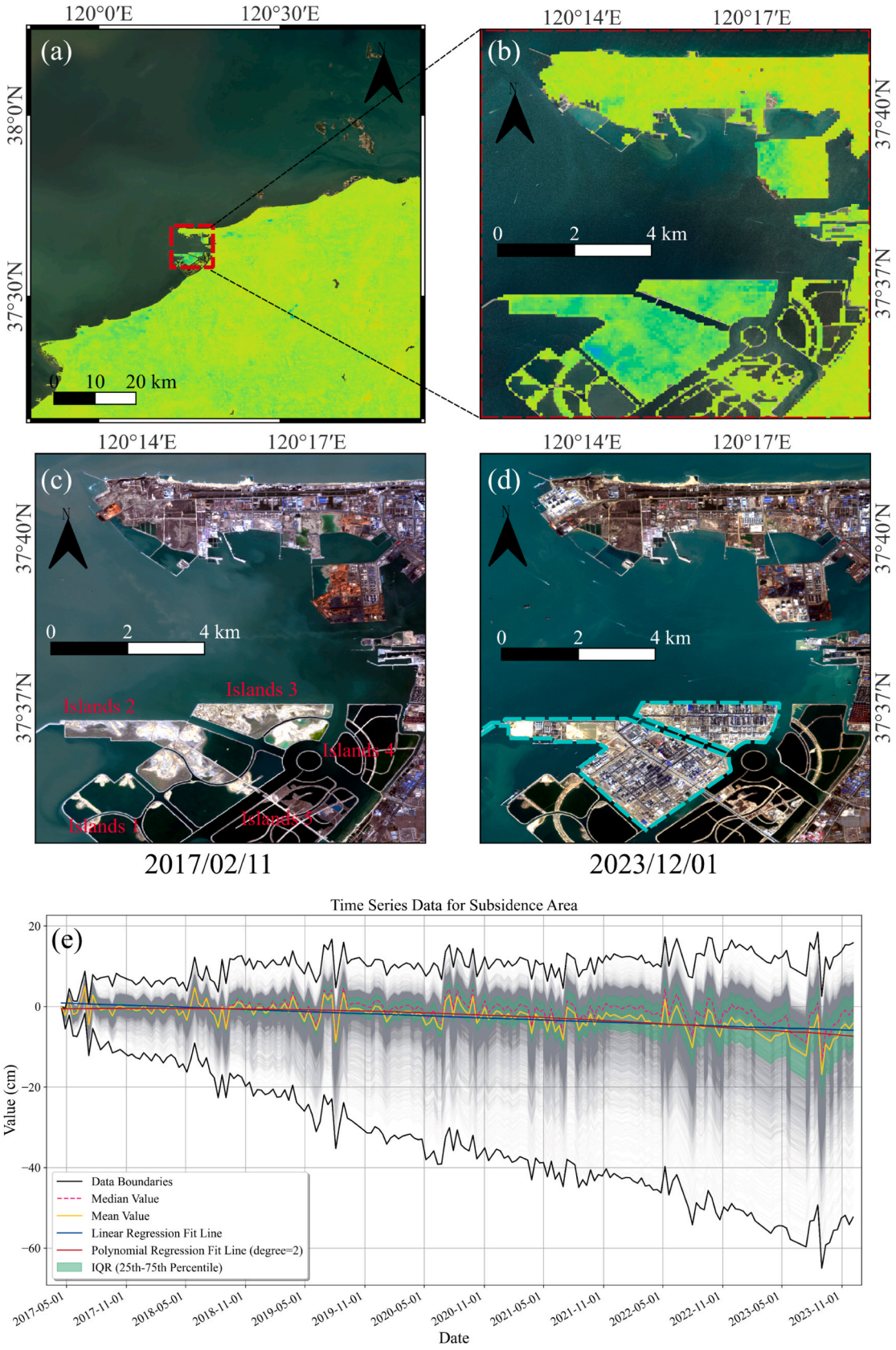


Fig. 10. (a)–(e) illustrate the spatial distribution of coastal subsidence for Zone C, along with the optical imagery and time series curves for the subsidence center. The green dashed polygon in the optical image of (d) represents the area with significant changes in surface features.

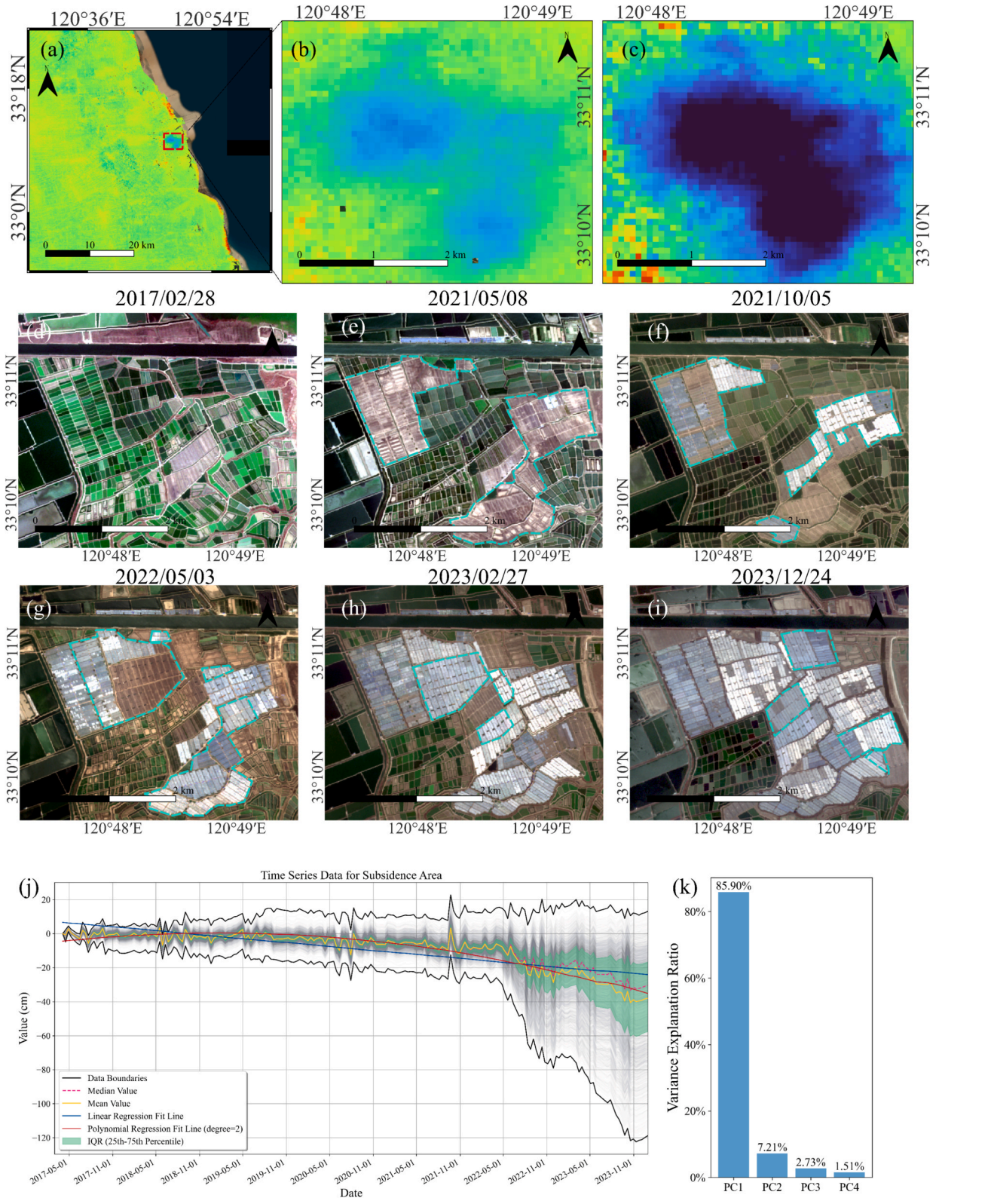


Fig. 11. (a)–(k) illustrate the spatial distribution of coastal subsidence for Zone D, along with the optical imagery, time series curves derived from all pixels of the subsidence center and corresponding PCA result. (a) and (b) show the average annual subsidence rate maps from 2017 to 2023, with (b) being a magnified view of the red dashed rectangle in (a). (c) shows the average annual subsidence rate for the same area as (b) during the period from April 2, 2022, to December 23, 2023. (d)-(i) are Sentinel-2 optical images of the area shown in (b) during six different time points, with the corresponding dates labeled above each image. (j) and (k) present the subsidence time series from 2017 to 2023 and the corresponding PCA results for the area shown in (b). The green dashed polygon in the optical image of (e)-(i) represents the area with significant changes in surface features.

5.2. Correlation between groundwater and subsidence

Land use classification serves as a direct indicator of human activities on the ground. Considering the land use categories can reduce the complexity and diversity of spatial drivers of deformation in typical areas, weaken the influence of other drivers, and to some extent, play a role in controlling variables (Wang et al., 2022a). Over time in Fig. 7, the magnitude of the PC1 show a similar nearly linear trend similar to subsidence, indicating that the drivers corresponding to the PC1 with VER > 95 % have a long-term linear dominant influence on subsidence. At the same time, it may have superimposed the seasonal variation signal of precipitation, causing the groundwater level to rise. The PC2 and PC3 are relatively stable and have a very small contribution (VER < 1 %) to subsidence. The PC4, as a high-order PC, has a variance contribution that can be ignored, mainly reflecting the influence of observational noise. It is worth noting that a sudden subsidence in Zone D related to the construction of greenhouse breeding ponds was observed, which led to significant fluctuations in the different PCs simultaneously. In addition, the deformation in areas with less subsidence is mainly affected by the elastic changes of the aquifer caused by seasonal precipitation and natural compaction (Huning et al., 2024; Peng et al., 2022; Shirzaei et al., 2021).

It can be seen that the significant coastal subsidence (>20 mm/yr) in the typical zones mainly results from the exploitation of underground resources, including brine and groundwater. This is consistent with 55 % of the published cases of land subsidence worldwide (Bagheri-Gavkosh et al., 2021; Huning et al., 2024). The existing global-scale studies on land subsidence from ground-based and space-borne monitoring techniques are usually based on discrete cases (Davydzenka et al., 2024; Hasan et al., 2023; Herrera-García et al., 2021; Huning et al., 2024; Nicholls et al., 2021; Nicholls and Shirzaei, 2024; Shirzaei et al., 2021). As far as we know at present, there are no cases covering both Shandong and Jiangsu provinces in eastern China.

Some studies further utilized the independent component analysis (ICA) to achieve independent signal separation based on PCA (Gualandi and Liu, 2021; Peng et al., 2022; Si et al., 2024). To perform the ICA, several fundamental assumptions must be satisfied: (1) the source signals must be statistically independent and non-Gaussian; (2) the observed signals should adhere to a linear mixing model; (3) the number of source signals cannot exceed that of the observed signals; (4) the mixing matrix must be invertible; (5) the mixing process should be noise-free or have negligible noise; and (6) the signals must be time-independent. Violation of these assumptions may significantly compromise the separation results. Furthermore, those research areas are relatively small in size and are mainly influenced by certain dominant factors, such as volcanic magma movement, periodic freezing and thawing in high-altitude regions, or groundwater extraction activities in inland areas (Dualeh and Biggs, 2025; Ebmeier, 2016; Gaddes et al., 2018; Martin et al., 2022). It is currently unclear whether this method remains effective in larger coastal areas where the ground has subsided due to heterogeneous causes (Gong et al., 2025; Wang et al., 2024c).

As shown in Fig. 12, we compared the spatiotemporal changes of groundwater level and vertical deformation. The location, monitoring layer, depth, and other detailed information of the groundwater monitoring wells are provided in **Table S3**. Considering the spatial distribution of wells and coastal subsidence, groundwater level and vertical deformation for areas with significant subsidence were selected for comparison. Since subsidence is a localized phenomenon rather than being confined to a single point, and the groundwater levels in the vicinity of a well are expected to be consistent, the average deformation of all InSAR data points within a 300-meter radius around each well was extracted.

Considering that both InSAR deformation and groundwater level time series exhibit certain periodic signals, especially the groundwater level may be related to seasonal recharge from precipitation, linear fitting was applied to both datasets to illustrate their respective trends,

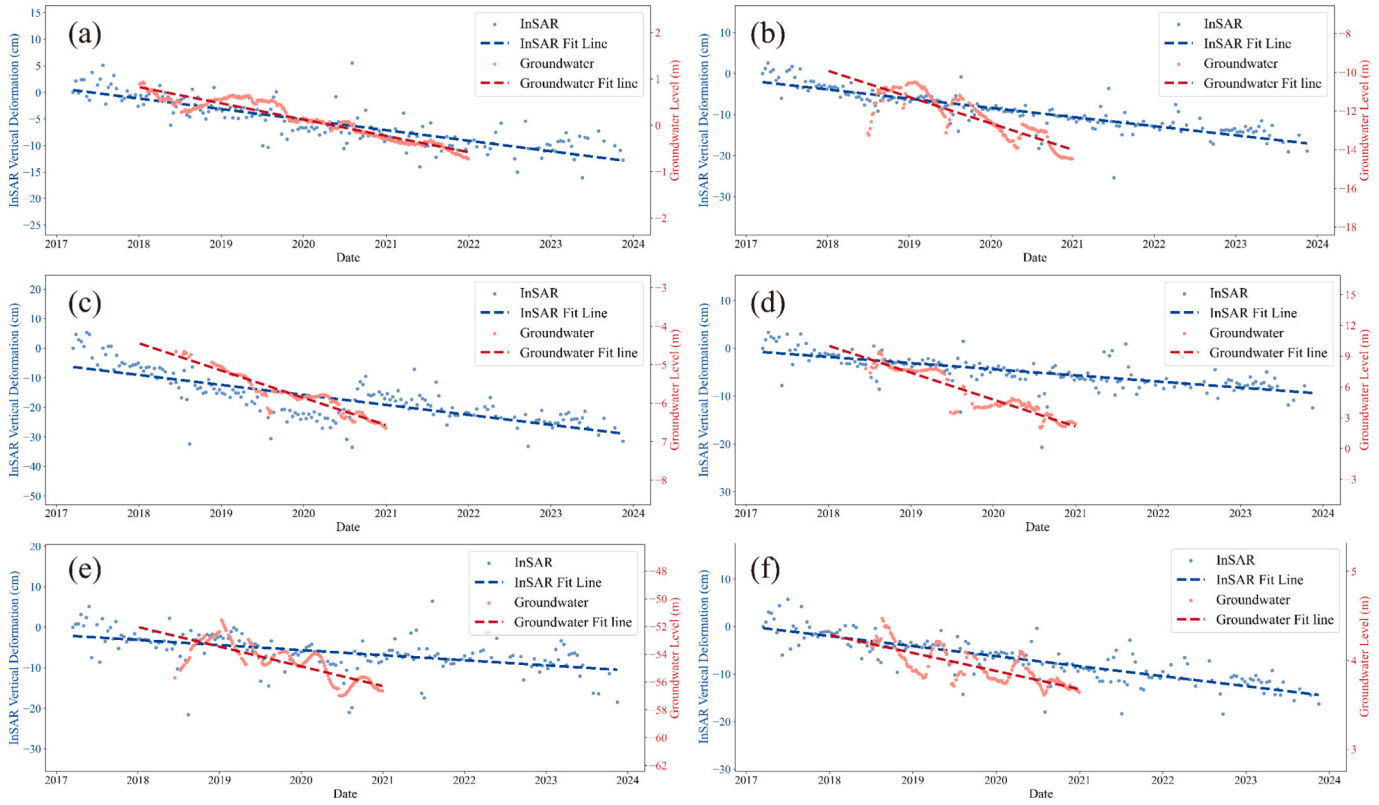


Fig. 12. Subplots (a)-(e) show the comparison between InSAR vertical deformation and groundwater level variations from six monitoring wells. The dashed lines represent their respective linear fitting lines. Panels (a)–(e) correspond to wells 1–6, respectively.

as shown by the dashed lines in Fig. 12. Although the groundwater level data only cover part of the InSAR observation period due to the limited time span of groundwater monitoring, the variation trend of groundwater level is highly consistent with that of coastal subsidence.

The fact is that the Quaternary sedimentary layers are prone to subsidence due to their extremely high compressibility (Haghshenas Haghighi and Motagh, 2024; Hasan et al., 2023; Huning et al., 2024; Nicholls and Shirzaei, 2024). When groundwater is excessively extracted, the water levels in the aquifers drop, reducing pore water pressure, which triggers subsidence due to the strata compression. Moreover, the coastal subsidence is largely unaffected by seasonal groundwater replenishment. Even if groundwater extraction were to cease, the subsidence may not immediately stop, but will likely continue for some time. Existing studies have shown that coastal subsidence amplifies relative sea level rise, which in turn can strongly exacerbate the flooding exposure of coastal residents (Ao et al., 2024; Fang et al., 2022; Huning et al., 2024; Minderhoud et al., 2025; Nicholls et al., 2021; Thiéblemont et al., 2024). Therefore, the government should take necessary measures for groundwater management as soon as possible, and continuously monitor the combined effects of coastal subsidence and sea level rise.

6. Conclusion

Coastal subsidence exacerbates relative sea-level rise, increasing the risks of coastal flooding and soil salinization. Despite ongoing research, the accurate spatial distribution and magnitude of coastal subsidence in eastern China remain poorly constrained, and the relative contributions of its primary drivers are not yet fully understood. In this study, we generated 90-meter-resolution TS-InSAR deformation rate maps using Sentinel-1 SLC data (2017–2024), covering the coastal regions of Shandong and Jiangsu provinces. The analysis revealed widespread hetergeneous subsidence, with pronounced subsidence funnels in northeastern Shandong and eastern Jiangsu. The maximum subsidence rate throughout the study period was 344 mm/yr, recorded in the YRD. In addition, regions experincing subsidence rates exceeding 20 mm/yr spanned 23,653 km^2, accounting for 15.6 % of the total study area.

By integrating Sentinel-2 imagery, subsidence time series data, and PCA, we characterized the subsidence dynamics in four typical regions with notable subsidence. Our analysis revealed strong correlations subsidence and groundwater level fluctuations. The results indicate that human activities, specifically underground brine extraction, groundwater pumping, and land reclamation, serve as the primary drivers of substaintial subsidence. Furthermore, our investigation shows that most subsidence funnels exhibit deformation patterns contorlled by a single dominant factor.

In the future, efforts should be made to strengthen the construction of an integrated monitoring network, as well as quantitative blind source separation algorithms and social and economic benefit assessment models. The management strategy should shift from a single engineering protection approach (e.g., the construction of seawalls) to a joint regulation of groundwater and surface loads to enhance the resilience of the coastal zone. This study is expected to provide critical insights for coastal authorities in effectively monitoring and mitigating subsidence risks, while establishing a robust methodological framework to guide future research endeavors.

CRediT authorship contribution statement

Peng Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jianbo Bai:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lin Shen:** Writing – review & editing, Methodology. **Wei Tang:** Writing – review & editing, Methodology. **Cunren Liang:** Writing – review & editing, Methodology. **Bin Zhao:** Writing – review & editing, Methodology. **Zhenhong Li:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Houjie Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jag.2025.105047.

Data availability

The coastal subsidence results are openly available through Zenodo platform (https://zenodo.org/records/17661835).

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